



Original Article

Isolation rearing does not constrain social plasticity in a family-living lizard

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An animal's social environment can be both dynamic and complex. Thus, social species often garner fitness benefits through being plastic in their social behavior. Yet, social plasticity can be constrained by an individual's experience. We examined the influence of early social environment on social behavior in the tree skink (*Egernia striolata*), a family-living lizard. In the first phase of this study, we reared juveniles in 2 different social environments for 1.5 years: either in isolation or in unrelated pairs. We quantified each lizard's sociability at 4-month intervals using a standardized laboratory assay and found that isolated lizards were more sociable, spending the assay closer to an adult female, than socially-reared lizards. In the second phase of this study (at the end of 1.5 years), we released all lizards into a semi-natural environment, observed their associations, and used social network analysis to quantify social behavior. During the initial 6 weeks post-release, we detected no differences in social behavior between rearing treatments. However, during the following 6 months differences emerged. Isolated lizards were more homogeneous in the strength of their associations than socially-reared lizards. Also, at first, isolated lizards associated more strongly than socially-reared lizards. Over time, isolated lizard associations became weaker and involved fewer lizards. In contrast, the level and number of associations of socially-reared lizards were stable over time. Our findings suggest that early experience influences tree skink social behavior but does not constrain social plasticity: isolation rearing did not limit their ability to respond to a novel social environment.

Key words: aggregation, developmental environment, plasticity, social competence, social network analysis, reptile.

INTRODUCTION

Social associations are influenced by resource availability, predation risk, mating system, parental care, and an individual's age and experience (Schutz et al. 2007; Ward and Webster 2016). These factors can interact in complex ways resulting in a dynamic social environment that is unpredictable, because it results from interactions between many individuals responding to inconstant environmental cues (Taborsky and Oliveira 2012). A social-living individual needs to constantly adjust its behavior in response to changes in their environment due to social and ecological factors, and this ability impacts their fitness (social plasticity; Stacey and Bock 1978; Oliveira 2012; Montiglio et al. 2017; Neelon and Höbel 2017). It is, therefore, expected that individuals of social species will be highly plastic in their social behavior and vary their

responses across social contexts to maximize individual fitness (termed “social competence”; Taborsky and Oliveira 2012).

The type and extent of social experience individuals are exposed to affects how they perceive, interpret, and act in social situations (Scott 1962; Taborsky and Oliveira 2012; Taborsky 2016; Bötting and von Engelhardt 2017). For example, isolation rearing negatively affects appropriate social behavior in mammals (Harlow et al. 1965; Toth et al. 2011), birds (Baron and Kish 1960), fishes (Hesse and Thünken 2014), and invertebrates (Liedtke and Scheider 2017; Schausberger 2017). To date, research investigating how early social experience affects social behavior has focused on obligate social animals with required parental care (e.g. mammals and birds; Baron and Kish 1960; Harlow et al. 1965; Mitchell et al. 1966; Varty et al. 2000). For example, Harlow's research on rhesus macaques (*Macaca mulatta*) in the 1960s often dominates our understanding of the impact social isolation can have on development, and may guide expectations that isolation severely negatively

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impacts an individual's behavioral development (Harlow et al. 1965; Mitchell et al. 1966). This focus has largely neglected the full spectrum of sociality, which includes species with only occasional interactions (e.g. interactions while defending territories or during mating) or those that form seasonal aggregations (Ward and Webster 2016). Recently, however, studies have begun to investigate the impact of social environment on behavioral development in solitary spiders with a transient social life-stage (*Agelena labyrinthica*; Lesne et al. 2016), and in squamate species for which social organization has not been studied in detail (Ballen et al. 2014; Hoss et al. 2015; Aubret et al. 2016).

Squamate reptiles as a group have traditionally been viewed as relatively asocial (Doody et al. 2012). Yet, there is evidence that social environment can influence reptile social behavior. For example, hatchling veiled chameleons (*Chameleo calyptratus*) reared in isolation are more submissive during conspecific interactions than socially-reared hatchlings (Ballen et al. 2014). Also, hatchling viperine water snakes (*Natrix maura*) incubated alone, versus in contact with other eggs, are less aggregative (Aubret et al. 2016), and depriving neonate cottonmouths (*Agkistrodon piscivorus*) of maternal attendance post-birth reduces their tendency to associate with conspecifics (Hoss et al. 2015). These studies suggest that the relationship between social experience and social competence found in other taxa might also apply to squamate reptiles; however, limited research on this topic prevents us from making generalizations.

Recent evidence suggests that sociality in lizards and snakes is much more common and varied than previously believed (Whiting and While 2017). Phylogenetic analysis shows that stable aggregations have evolved multiple times in squamates, and, of these aggregations, a small proportion live in stable family groups (Gardner et al. 2016). The tree skink (*Egernia striolata*) aggregates within tree hollows or crevices, and cracks in rocks (Cogger 2014). This species can be found alone, or within groups consisting of adult pairs with or without offspring, or only of juveniles (Bonnett 1999; Duckett et al. 2012). Group size varies from 2 to 10 individuals, and there is within- and between-population variation in tree skink social behavior (Bustard 1970; Bonnett 1999; Duckett et al. 2012). This variation in group size and social behavior allows investigation of the influence of social environment on individual tree skink behavioral development and social behavior as adults, which we did in an experimental setting.

We examined how the social environment during early life affects tree skink social behavior. In the first phase of the study, we reared lizards within one of 2 social environments (isolation or within a pair) for 1.5 years, and quantified juvenile sociability with a laboratory assay. In the second phase, we released lizards into a semi-natural environment and recorded their association patterns over the short- (initial 6 weeks) and long-term (subsequent 6 months: weeks 7–34). We hypothesized that isolation rearing would reduce lizard social competence, and predicted that isolated lizards would avoid aggregating with conspecifics, as has been observed in other squamates, fishes, invertebrates, and mammals (Harlow et al. 1965; Hesse and Thünken 2014; Aubret et al. 2016; Schausberger 2017). An individual's social competence is also reflected in its ability to change social behavior across situations (Taborsky and Oliveira 2012). In this regard, the social environment during development can constrain an individual's behavioral plasticity (Harlow et al. 1965; Taborsky et al. 2012; Hesse and Thünken 2014). To investigate this, we quantified consistency in an individual lizard's social behavior to examine plasticity in these traits and whether consistency differed between rearing treatments and contexts (Aplin et al. 2015; Stamps 2015).

METHODS

In this experiment, we used 66 tree skinks that were offspring from 35 females collected near Albury, New South Wales, Australia (35.98°S, 146.97°E). These offspring were from 2, yearly cohorts (2014 and 2015; see [Supplementary Material](#) for details on captive husbandry and housing). The data from the laboratory portion of this experiment has been previously published in Riley et al. (2017b) in which we quantified 4 behavioral traits (exploration, boldness, sociability, and aggression) of tree skinks throughout development. This current study aims to examine how social environment affects tree skink social behavior. We re-analyzed the Riley et al. (2017b) sociability data (see Statistical Analyses section below) and changed our focus to comparing sociability between rearing treatments (isolation and social). This allowed us to interpret the adult social behavior of isolated and socially-reared tree skinks that we quantified in semi-natural enclosures (the second and novel part of our study; see below), in relation to the treatment differences in sociability during the juvenile life-stage (Riley et al. 2017b).

Rearing treatment and laboratory sociability assays

After lizards were born (within a maximum of 12 h), we separated them from their mothers and housed them individually (for dates of parturition see [Supplementary Material](#)). After all juveniles were born each year, we conducted the first sociability assay (see below).

After the first assay, we then randomly allocated juveniles into 2 social environments: isolated (housed alone; $N_{2014} = 14$ lizards and $N_{2015} = 16$ lizards), and social (2 unrelated juveniles housed together; $N_{2014} = 14$ lizards within 7 pairs and $N_{2015} = 22$ lizards within 11 pairs). We split the number of offspring from each mother across treatments (see [Supplementary Material](#) for details), but were unable to sex juveniles prior to treatment allocation so we accounted for sex in our statistical analyses. Tree skink social groups often consist of parent(s) and offspring (Chapple 2003), but we were unable to recreate this environment in captivity due to the risk of infanticide (Lanham and Bull 2000; Post 2000; O'Connor and Shine 2004; Riley JL, unpublished data). The group size we used in our experiment is present in wild tree skink populations, because juvenile-only group size ranges from pairs to 4 individuals and juveniles are also observed on their own (Bonnett 1999; Michael and Cunningham 2010; Duckett et al. 2012; Riley JL, unpublished data).

Once within their treatments, we assayed lizard sociability 3 more times: at ca. 5, 7, and 12 months of age (see [Supplementary Material](#) for exact dates). We measured sociability within 2 batches due to space limitations of our experimental room (maximum of 24 lizards per batch ranging from 13 to 16 days apart; see [supplementary materials](#) for exact dates).

During sociability assays we individually housed lizards in opaque, plastic arenas, which were separated into 2 compartments with a clear, Perspex® divider placed 11 cm from one end (Riley et al. 2017b; see [Supplementary Material](#) for more details). We first placed the focal juvenile within a shelter in the larger compartment and then placed an unrelated, adult female in the smaller compartment on the opposite side of the divider. Females were unrelated and unfamiliar to the focal lizard. We randomly paired females with each focal lizard, and ensured the female was different in each repetition of the assay (i.e. used only once/lizard). We then lifted the juvenile's shelter and remotely video-recorded the location of the juvenile in relation to the female for the next 5 h. From the

video, at 10 min intervals over the full course of the trial, the juvenile was scored as being within one of 4 lateral quadrats (11 cm width) that sequentially reflected distances further away from the female. From these data, we calculated the weighted mean distance the juvenile was from the female across the whole trial, by multiplying the number of times in each quadrat (Q1, Q2, Q3, and Q4) by the mean distance the quadrat was away from the female's compartment (5.5 cm, 11 cm, 16.5 cm, and 22 cm, respectively) and dividing the product by the total number of observations ($N_{\text{obs}} = 30$). This weighted mean was used to quantify sociability across the first year of life; lower values reflect higher sociability. Due to issues with some videos (i.e. failure to record particular trials), 5 juveniles were removed from this portion of our analysis: the final sample size of our laboratory sociability assay was 26 juveniles in 2014 (14 socially-reared and 12 isolated), and 35 juveniles in 2015 (20 socially-reared and 15 isolated).

Spatial associations

We quantified lizard sociability within a semi-natural environment after they reached adulthood to assess long-term effects of rearing treatment on social associations. Tree skinks reach sexual maturity after ~1.5 years in captivity (Riley JL, unpublished data). We used all 28 lizards reared in 2014 in this experiment (14 isolated and 14 socially-reared), and released them into a fenced outdoor enclosure with an area of 70 m² (10.5 × 6.7 m) on 14 September 2015. A net surrounded the enclosure to exclude avian predators (Supplementary Figure S1).

We provided 28 artificial crevices (hereafter refuges) inside the enclosure. This ensured that refuges were not a limited resource, thus allowing aggregation to depend on individual preference and not ecological factors (i.e. a limited resource). We placed 2 roofing tiles (each tile was 410 × 260 mm) on top of each other to form refuges (Supplementary Figure S1). These refuges were placed in 4 rows of 7 (0.75 m apart and 1.25 m away from the perimeter; Supplementary Figure S1). Prior to release, we permanently marked all lizards with microchips (PIT tags) and visually numbered them with 3 circles of numbered cloth tape (Tesa®, Hamburg, Germany; Supplementary Figure S1C; Olsson and Shine 2000; While et al. 2009a). The lizards were all released in the middle of the enclosure. If a lizard lost any marking tape, it was re-marked at the end of the day, after data collection, and immediately released back into its refuge of capture.

We recorded lizard spatial positions within the enclosure over 2 time periods: the initial 6 weeks post-release (4× daily) and then the following 6 months (weeks 7–34; twice weekly). Observers (C.G. and S.T.L. during the initial 6 weeks; C.G. and C.F. for the following 6 months) were blind to rearing treatment. Data collection took the same amount of time in both periods, on average 22 min (95% CI = 21 to 23), which reflects similarity in sampling intensity and effort. Our sampling methods (see below) differed between 6-week and 6-month observations, and we quantified different social behaviors. Our short-term observations captured behavior across the daily activity period of lizards, whereas the long-term observations recorded which lizards were sharing refuges overnight and social associations before the start of daily activity.

During the initial 6 weeks (14 September to 25 October 2015), we recorded lizard spatial position within the enclosure 4 times each day at 10:00, 12:00, 14:00, and 16:00 h. We recorded locations for all lizards that we could visually observe (see Results for observation rates). First, we visually scanned the entire enclosure from a raised tower (2.1 m tall), and then we slowly walked around the outside

of the perimeter for a ground-level perspective thereby maximizing our ability to detect lizards. After the initial 6 weeks, we modified our protocol for the following 6 months (29 October 2015 to 29 April 2016) to focus on lizard social associations overnight (i.e. during shelter in refuges) rather than those during their daily activity period. We recorded lizard spatial positions every Monday and Friday, between 07:00 and 09:00 h, prior to the tree skink's activity period. At that time of day, lizards were still sheltering within refuges. During these observations we lifted each roofing tile, captured every lizard in the refuge to identify them from their PIT tag number using a PIT tag reader (lizards did not retain their identifying stickers during the 6-month observations), and then placed them back at their site of capture. We also checked under water dishes and in the area between refuges for lizards. Each sampling period was started from a different location, and performed in a different pattern during each trial.

Statistical analyses

Analyses of laboratory sociability scores

We examined if tree skink sociability differed between rearing treatments, using a linear mixed effects model (LMM, *lmer* in the *lme4* R package; Bates et al. 2015; R Core Team 2016). In our model, we examined if the response variable, *sociability* (the weighted mean distance (mm) a lizard was located from an adult female during a trial), was affected by *rearing treatment* (isolated or social) while controlling for the additional fixed factors of *age* (continuous), *sex* (male or female), *cohort* (2014 or 2015), *batch* (1 or 2), *body temperature* (continuous), and *body condition index* (continuous; residuals from a simple linear regression between log-transformed mass and snout–vent length). Continuous fixed factors were z-transformed (mean centered and divided by standard deviation) before analysis, which standardizes the variables and facilitates interpretation of main effects in the presence of interactions (Schiegg 2010). To control for dependencies within our data from sampling each lizard repeatedly we included a random intercept and slope for lizard identity across age. To control for repeated sampling of individuals from the same litter, we also included a random intercept for mother identity. We ensured that there were no influential outliers, no strong collinearity, and that the model assumptions of normality of residuals and homogeneity of variance were met (as per Zuur et al. 2010; see Supplementary Materials for details).

We also examined the consistency of an individual's sociability and determined if this was affected by rearing treatment. To accomplish this, we calculated adjusted repeatability ($R_{\text{adj}}|\text{age}$; Biro and Stamps 2015) for each treatment and the associated 95% confidence intervals by bootstrapping the data 1000 times with the *boot* function from the R package *boot* (Davison and Hinkley 1997; Canty and Ripley 2017), while controlling for the same covariates that were within our LMMs (Nakagawa et al. 2010; Biro and Stamps 2015). We compared $R_{\text{adj}}|\text{age}$ between treatments by examining the overlap of each rearing treatment's 95% CIs. Repeatability ($R_{\text{adj}}|\text{age}$) theoretically ranges between 0 (individuals never express the same trait value over repeated measures) and 1 (individuals always express the same trait value over repeated measures; Nakagawa et al. 2010). Yet, in practice, the average repeatability observed in studies on animal behavior is 0.37 (Bell et al. 2009).

Social network analysis

Animal social network analysis is a powerful technique for quantifying association or interaction data (Farine and Whitehead 2015).

We used social network analysis to quantify the associations we observed between lizards in the semi-natural enclosure. We considered lizards to be associating when they were at the same refuge within a sampling period (i.e. if 2+ lizards were located in/on the same refuge, they were defined as a group). Additionally, the refuges in our study were small (e.g. 410×260 mm; approximately the size of 2 lizards), thus if lizards were sharing a refuge they would have been aware of each other. Association strength for each pair of lizards (dyad) was calculated using the half-weight association index (HWI). The HWI ranges between 0 (never observed in the same refuge) and 1 (always observed in the same refuge). It is a relative measure of association strength, which is most appropriately used when an entire population cannot be observed during each sampling period (Cairns and Schwager 1987), as is the case in our study (see Results for sampling rate).

We constructed 6 weekly networks and 6 monthly networks (i.e. one for each of the 6 weeks, and one for each of the next 6 months) that controlled for lizard space use (i.e. the function included refuge number, which reflects lizard location, during construction of the group-by-individual matrix; Farine 2013) to ensure the associations we were describing were based on social, and not abiotic, biotic, or spatial factors (Figure 1). The edge weights in these networks were the HWI association strength (described above), resulting in weighted, undirected networks in which individuals (nodes) were connected if $\text{HWI} > 0$. Network construction and analysis were performed using the *asnipe* R package (Farine 2013).

For each network (6 weekly and 6 monthly networks), we calculated 3 network metrics that quantified individual social behavior: binary degree, weighted degree, and the coefficient of variation (CV) of edge weights. Binary degree is the number of lizards the focal lizard was observed sharing a refuge with, and weighted degree (also termed “strength”; Whitehead 2008) is the sum of edge weights (dyadic HWIs) of the focal lizard (Whitehead 2008). These metrics both reflect the sociability of an individual and are complementary; an individual can have high numbers of associates with weak associations (high binary degree and low weighted degree) or, in contrast, a low number of strong associations (low binary degree and high weighted degree). The CV of edge weights

is a measure of social heterogeneity (also termed “social differentiation” and the “clustering coefficient”; Whitehead 2008) of a focal lizard and quantifies variability of an individual’s relationships (Leu et al. 2016). High values of social heterogeneity means that relationships are variable, and that the focal lizard is mainly associating strongly (i.e. preferentially) and/or weakly (i.e. avoiding) with conspecifics (Farine and Whitehead 2015). In contrast, low values of social heterogeneity means that relationships are more homogeneous, and that focal lizards are associating relatively evenly with conspecifics (Farine and Whitehead 2015).

Temporal aspects of social relationships

We used separate LMMs to determine if the observed network metrics (binary degree, weighted degree, and CV of edge weights) differed between rearing treatments across each time period (6 weeks or 6 months). All LMMs included the fixed factors of *time period* (week or month, respectively), *rearing treatment* (isolated or social), *sex* (male or female), as well as interactions between *time period* and *sex* and *time period* and *treatment*. If interactions were not significant (according to P_{rand} , see below) they were removed and the models re-fitted. Models also included the random intercept and slope of lizard identity across time, and the random intercept of mother identity. Our observed network metric dataset was explored prior to statistical analyses to investigate normality, the presence of outliers, and collinearity. Additionally, the assumptions of normality of residuals and homogeneity of variance were verified for all LMMs analyzing observed data (Zuur et al. 2009; see supplementary materials for details). Binary and weighted degree were $\log(x+1)$ transformed to normalize the data. We also wanted to examine the consistency of social metrics for each rearing treatment, and calculated treatment-specific $R_{\text{adj}} | \text{time}$ (weeks or months; Biro and Stamps 2015) using the same protocol as described above.

Hypothesis testing for data generated from social networks is based on null models constructed with data from random permutations (Farine and Whitehead 2015). This is necessary because of non-independence of the data from social networks. Permutations were done using our group-by-individual matrix by randomly swapping individuals between groups, while controlling for location

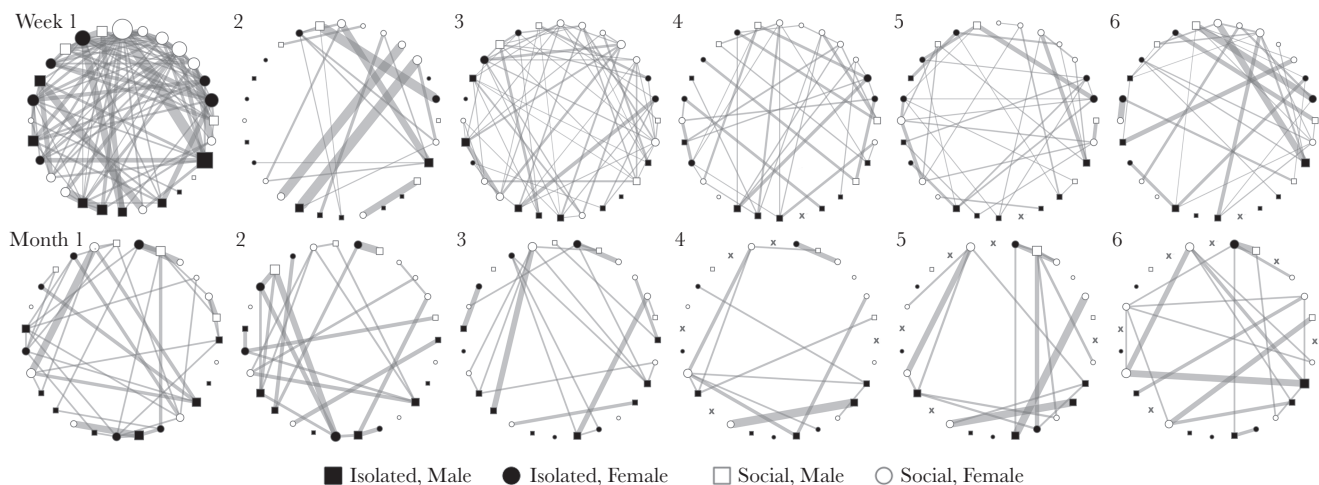


Figure 1

Weighted social networks across our 6-week (top) and 6-month observations (bottom). We aggregated the association data for each week or month, respectively, and constructed 6 weighted, undirected networks for each time period. These social networks reflect the association matrices used in our analyses. Node size reflects weighted degree of each individual. If lizards were removed from the social network during a time period (e.g. due to death or injury), nodes are replaced with a grey “X”.

because original network construction already accounted for this (Farine 2013). These permutations were done separately for each weekly/monthly network. Importantly, this permutation technique retains the structure of our observed dataset: it maintains the same number of dyads observed, number of times an individual is sighted, and number of individuals recorded during each sampling period as our observed data (Whitehead 2008; Croft et al. 2009; Farine and Whitehead 2015). During the first 6 weeks, 1 lizard (female, socially-reared) died of natural causes, and during our 6-month observations, predators unexpectedly infiltrated the enclosure resulting in the removal of 5 lizards (3 isolated males, 1 isolated female, 1 socially-reared male) during the third month, and 1 lizard (socially-reared female) during the fifth month due to predation and/or injury. Our permutation technique took this into consideration.

From the randomized data, we then reconstructed the networks, derived the same 3 social metrics, and conducted the same LMMs and calculations of $R_{adj}|time$ as we did for the observed data. Randomizations were repeated 10000 times, and P values (P_{rand}) for each effect were calculated by comparing model coefficients from the observed data to the distribution of model coefficients based on the randomized data (Aplin et al. 2015; Farine and Whitehead 2015; Leu et al. 2016). We considered effects to be significant if observed values fell outside the 95% range of the random coefficient distributions. We compared $R_{adj}|time$ between treatments by examining overlap of each rearing treatment's 95% CIs. We assessed if observed $R_{adj}|time$ differed from what you would expect by chance alone by examining overlap between 95% CIs for observed $R_{adj}|time$ estimates and the 95% range of the random $R_{adj}|time$ estimates.

RESULTS

Laboratory-based sociability assays

Socially-reared lizards were located further from an adult female, exhibiting lower sociability, than isolated lizards (Table 1). Distance from an adult female decreased as lizards aged, reflecting increasing sociability, for both isolated and socially-reared lizards as they aged (Table 1). There were no batch, cohort, or sex effects on

Table 1

Effects of rearing treatment (ISOLATED or SOCIAL) and age on lizard sociability ($N_{obs} = 244$, $N_{juv} = 61$, $N_{mom} = 35$) that was quantified using a laboratory behavioral assay

Fixed effects	<i>B</i>	<i>SE</i>	<i>t-value</i>	<i>P</i>
Intercept (SOCIAL, 1, 2014, and FEMALE)	16.636	0.769	21.645	<0.001
Age	-0.939	0.229	-4.099	<0.001
Rearing treatment (ISOLATED)	-1.459	0.727	-2.007	0.045
Batch (2)	-0.521	0.402	-1.297	0.195
Cohort (2015)	-0.769	0.833	-0.923	0.356
Body temperature	0.307	0.194	1.581	0.114
Body condition index	0.374	0.203	1.837	0.066
Sex (MALE)	0.188	0.798	0.236	0.813
Random effects	σ^2			
Juvenile identity	5.237			
Maternal identity	1.573			
Residual	6.792			

Significant coefficients are bolded. The LMM formula in R was `lmer(sociability ~ age + social_treat + batch + cohort + body_temp + body_cond + (1+age|liz_id) + (1|mom_id))`. The interaction between age and rearing treatment was not significant, so it was removed and the model re-fit.

the distance lizards were located from an adult female, and this distance was also not related to body temperature or body condition (Table 1). $R_{adj}|age$ of sociability was moderate (isolated: $R_{adj}|age = 0.427$, 95% CI = 0.216 to 0.637; social: $R_{adj}|age = 0.304$, 95% CI = 0.000 to 0.665), and did not differ between rearing treatments (Figure 2A).

Social relationships in a semi-natural environment

During the initial 6 weeks post-release, all 28 lizards were observed at least once during 168 sampling periods (6 weeks $\times$ 7 days $\times$ 4 observations per day; total observations of lizards = 2061). Each sampling period 48% (95% CI = 44 to 52) of the lizards were observed; but each week, across 28 sampling periods, 98% (95% CI = 95 to 100) of the lizards were observed. In the following 6 months, all 27 lizards were observed at least once during a total of 52 sampling periods (28 weeks $\times$ 2 observations per week; total observations = 985). Within each sampling period during these 6 months, 98% (95% CI = 97 to 100) of the lizards were observed.

During both time periods, mean group size was 2 (6 week: standard deviation = 0.28, range = 2–8; 6 month: standard deviation = 0.28, range = 2–4; Figure 1). However, individuals were also frequently observed alone (Figure 1). Lizards were observed alone in a refuge 80.4% (1340/1667) of the time during the initial 6 weeks, and 77.3% (612/792) of the time during the next 6 months. Thus, our network metrics and model parameter estimates that quantify social associations are lower than would be expected in a species that constantly associates with other individuals (Table 2).

Temporal variation in social relationships

Initial 6 weeks

Network metrics (binary degree, weighted degree, and CV of edge weights) did not differ between rearing treatments during this period (Table 2A). Sex did not affect binary degree or CV of edge weights, but female lizards initially had stronger associations (higher weighted degree) than males and decreased in the strength of their social associations (weighted degree) more quickly over time than males (Table 2A). In general, the number and strength of associations (binary and weighted degree) decreased over time, whereas social heterogeneity (CV of edge weights) increased over time (Table 2A; Figure 2).

$R_{adj}|week$ of our 3 social metrics were low for both rearing treatments. $R_{adj}|week$ did not differ between rearing treatments, and was not different from what we would expect by chance alone (95% CIs from both the observed and random $R_{adj}|week$ overlap). Nonsignificance was likely due to large variation in our observed $R_{adj}|week$ estimates (Fig. 2B).

Subsequent 6 months

Initially, isolated lizards had stronger associations (higher weighted degree) than socially-reared lizards (Table 2B). Over time, isolated lizards decreased in their number of associates (binary degree) and their associations became weaker (weighted degree). In contrast, socially-reared lizards were constant in their number of associates and level of associations (binary and weighted degree) over time (Table 2B). Social heterogeneity (CV of edge weights) was significantly lower in isolated than socially-reared lizards, and social heterogeneity decreased over time in both rearing treatments (Table 2B; Figure 1). None of the network metrics were affected by sex (Table 2B; Figure 1).

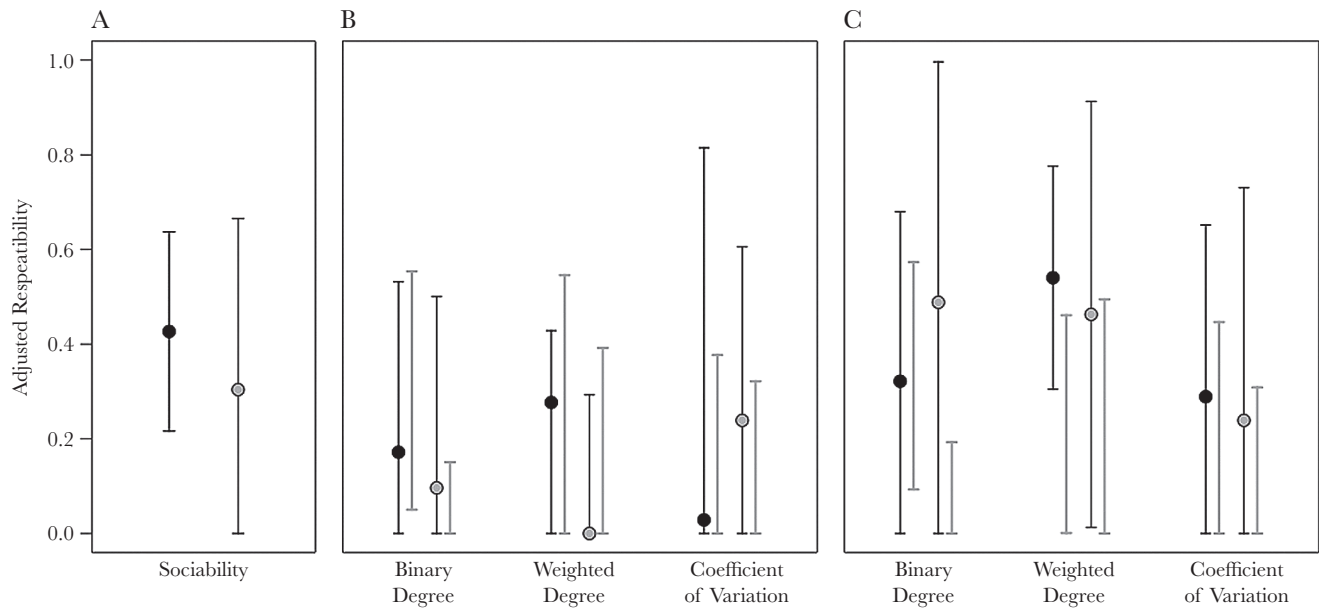


Figure 2

Adjusted repeatability ($R_{adj|time}$) and 95% confidence intervals for isolated (black circle and lines) and socially-reared (grey circle and black lines) lizards for (A) laboratory sociability scores, and the 3 individual network metrics: binary degree, weighted degree, and coefficient of variation across the (B) initial 6 weeks and (C) subsequent 6 months. $R_{adj|time}$ was not significantly different between rearing treatments, because 95% CIs for both rearing treatments (black error bars) overlapped. Grey bars show the 95% range of the $R_{adj|time}$ estimates calculated from 10000 data randomizations controlling for location. Our observed $R_{adj|time}$ was not different than what you would expect from chance alone, because the 95% range of the random $R_{adj|time}$ estimates (grey bars) overlapped with 95% CIs for observed $R_{adj|time}$ estimates (black error bars) in all cases.

$R_{adj|month}$ did not significantly differ between rearing treatments, because 95% CIs overlapped (Figure 2C). $R_{adj|month}$ of binary degree, weighted degree, and CV of edge weights ranged between moderate to high for both rearing treatments (Figure 2C). Our observed $R_{adj|month}$ did not significantly differ from what was expected by chance alone (95% CIs from both the observed and random $R_{adj|month}$ overlap; Figure 2C).

DISCUSSION

We generally found a positive relationship between isolation rearing and social associations in tree skinks, in contrast to our prediction. During the juvenile life-stage, isolated lizards spent laboratory trials closer to an unrelated adult female than socially-reared lizards. After lizards reached adulthood we released them into a semi-natural enclosure, and, after the first 6 weeks post-release, we found isolated lizards initially associated more strongly with others than socially-reared lizards (weighted degree). Isolated lizards were also more homogeneous in their social associations than socially-reared lizards (i.e. lower CV of edge weights). Interestingly, over the 6-month period, isolated lizards gradually began to associate with fewer lizards, more weakly (i.e. decreased in binary and weighted degree), whereas socially-reared lizards were stable in associations over time. This suggests that individuals reared in isolation were able to respond flexibly to their new social environment. In further support of this conclusion, our repeatability ($R_{adj|time}$) estimates for isolated lizards' social behavior were low to moderate. However, repeatability did not differ between rearing treatments, nor from what is expected by chance alone. This suggests that tree skink social behavior is impacted by isolation rearing, and also that, regardless of isolation rearing, tree skinks maintain plasticity in social behavior.

Sociability during the juvenile life-stage

Isolation rearing increased juvenile affiliation with an adult female. This finding was the opposite of what we had hypothesized: that isolation would result in individuals that avoid social situations and/or exhibit costly social behavior (e.g. high aggression; Harlow et al. 1965; Mitchell et al. 1966; Hesse and Thünken 2014; Leidtke and Schneider 2017; Schausberger 2017). Our prediction was based predominately on studies of species that have obligate parental care like mammals and birds (Baron and Kish 1960; Harlow et al. 1965), where socially manipulating these species during development results in costly physiological (e.g. abnormal levels of sex and stress hormones; Kaiser and Sachser 2005; Bölting and von Engelhardt 2017) and behavioral changes that reduce fitness (e.g. avoidance of conspecifics, inappropriate mating behavior; Harlow 1965; Yu et al. 2013; Hampson and Schiwitz 2016). Tree skinks' rudimentary parental care and facultative social system differs from that of mammals and birds. Therefore, we need to consider how our findings may influence tree skink fitness, while considering their social system.

Social associations, including affiliative ones, are not always beneficial and can be costly and even fatal in some cases. For example, yellow-bellied marmots (*Marmota flaviventris*) that are more affiliative are more likely to die during hibernation, potentially because hibernating in close proximity to other individuals may disrupt required thermoregulation (Yang et al. 2017). Although we did not directly quantify the fitness consequences of the altered social behavior we observed in isolation-reared tree skinks, we interpret our findings based on our knowledge of *Egernia*-group social systems (While et al. 2015; Whiting and While 2017). In the wild, unrelated adult *Egernia* spp. pose a direct mortality threat to juveniles; they are often highly aggressive and infanticide can occur (*E. stokesii*, Lanham and Bull 2000; *E. hosmeri*, Post 2000; *E. saxatilis*;

Table 2
Effects of sex (MALE or FEMALE), rearing treatment (ISOLATED or SOCIAL), and time (either week or month) on individual network metrics

(A) Short-term data (6 weeks; $N_{obs} = 168$, $N_{gr} = 28$, $N_{mm} = 15$)

	Binary Degree			Weighted Degree			Coefficient of Variation		
	β_{obs}	β_{rand}	P_{rand}	β_{obs}	β_{rand}	P_{rand}	β_{obs}	β_{rand}	P_{rand}
Intercept (FEMALE, SOCIAL)	1.682 (1.406, 1.958)	1.775 (1.639, 1.918)	0.901	0.671 (0.543, 0.799)	0.376 (0.337, 0.426)	<0.001	215.194 (153.606, 276.782)	221.103 (191.002, 250.109)	0.632
Week	-0.010 (-0.151, -0.042)	-0.059 (-0.085, -0.038)	0.002	-0.083 (-0.112, -0.056)	-0.050 (-0.061, -0.041)	<0.001	22.942 (7.457, 38.428)	6.010 (-0.362, 13.146)	<0.001
Sex: MALE	-0.087 (-0.348, 0.174)	0.043 (-0.077, 0.155)	0.230	-0.133 (-0.322, 0.056)	-0.022 (-0.114, -0.065)	0.006	13.793 (-29.213, 60.799)	4.227 (-24.644, 31.422)	0.323
Social Treatment: ISOLATED	-0.012 (-0.233, 0.210)	0.034 (-0.081, 0.136)	0.869	-0.010 (-0.092, 0.072)	0.003 (-0.017, 0.021)	0.357	-19.010 (-63.555, 25.535)	-10.786 (-38.004, 18.176)	0.310
Week*Sex	—	—	—	0.024 (-0.024, 0.072)	0.007 (-0.013, 0.028)	0.039	—	—	—
Week*Social Treatment	—	—	—	—	—	—	—	—	—

(B) Long-term data (6 months; $N_{obs} = 162$, $N_{gr} = 27$, $N_{mm} = 15$)

	Binary Degree			Weighted Degree			Coefficient of Variation		
	β_{obs}	β_{rand}	P_{rand}	β_{obs}	β_{rand}	P_{rand}	β_{obs}	β_{rand}	P_{rand}
Intercept (FEMALE, SOCIAL)	0.953 (0.601, 1.304)	1.938 (1.781, 2.090)	1.000	0.307 (0.186, 0.428)	0.210 (0.173, 0.250)	<0.001	369.011 (274.825, 463.197)	222.483 (193.372, 256.836)	<0.001
Month	-0.035 (-0.139, 0.068)	-0.105 (-0.140, -0.066)	1.000	-0.011 (-0.041, 0.019)	-0.012 (-0.021, -0.004)	0.532	-19.566 (-40.925, 1.793)	6.066 (-1.644, 13.775)	<0.001
Sex: MALE	-0.028 (-0.304, 0.248)	0.033 (-0.097, 0.156)	0.679	-0.032 (-0.144, 0.079)	-0.006 (-0.041, 0.031)	0.095	-11.264 (-101.847, 79.319)	7.326 (-21.646, 35.802)	0.502
Social Treatment: ISOLATED	0.486 (0.017, 0.955)	-0.284 (-0.534, 0.024)	0.067	0.155 (-0.009, 0.319)	-0.030 (-0.039, 0.108)	<0.001	-50.382 (-137.910, 37.147)	-12.225 (-40.907, 17.167)	0.002
Month*Sex	—	—	—	—	—	—	—	—	—
Month*Social Treatment	-0.144 (-0.288, -0.000)	0.091 (0.029, 0.148)	0.039	-0.044 (-0.086, -0.002)	-0.015 (-0.031, -0.000)	<0.001	—	—	—

Coefficients and 95% confidence intervals (italics in brackets) are presented for both observed and randomized data sets. P_{rand} is also presented, which is the comparison between the coefficients from observed data (β_{obs}) to the distribution of coefficients from the randomized data (β_{rand}). Effects are considered significant if observed coefficient values are outside the 95% range of random coefficient distributions; we have bolded these significant effects. The LMM formula in R was lmer(binary_degree ~ week + sex + social_treat + sexweek + social_treatweek + (1+week|ltx_id) + (1+week|ltx_id)), and respectively for association strength and coefficient of variation. If interactions were not significant (according to P_{rand}), they were removed and the models re-fitted.

O'Connor and Shine 2004; *Liopholis whitii*, Sinn et al. 2008; While and Wapstra 2008). In a previous study, we quantified the nature of the interactions that occurred between individuals that were reared in pairs (Riley et al. 2017b). These individuals were also used in this experiment. We found that socially-reared lizards experienced aggressive interactions within their social pair (Riley et al. 2017b), which influenced their growth rate and behavioral development. These aggressive interactions between juveniles may have exposed socially-reared lizards to a learning experience that facilitated anticipation of potentially dangerous interactions with conspecifics. Within our laboratory sociability assay, avoiding the unfamiliar, unrelated adult, as socially-reared lizards did, may be the most beneficial behavior to exhibit. In contrast, isolation rearing resulted in socially naïve lizards with greater affiliation towards unfamiliar, adult females; we hypothesize that this is a costly and maladaptive behavioral response.

Further research is needed to examine this hypothesis, as well as the fitness and developmental implications (i.e. reproductive success, survival, and neurological consequences) of social rearing environment in tree skinks, and other facultatively social animals. Many facets of development in obligate social species are affected by social rearing environment, which has long-term fitness implications (Mason and Sponholz 1963; Hampson and Schwitzer 2016). Yet, as we show in this study, the impact of social environment on development of facultative social species is not always consistent with findings in obligate social species. For example, the cognitive ability of obligate social species is often hindered by isolation rearing (e.g. primates, Harlow et al. 1965; rats, Amitai et al. 2014). In contrast, tree skink cognitive ability was unaffected by social rearing environment (i.e. spatial learning ability; Riley et al. 2017a and associative and social learning ability; Riley et al. 2018a). This lack of consistency in the relationship between social rearing environment, development, and long-term fitness across species differing in sociality, suggests each social system has unique set of selective pressures, and highlights the need for further research.

Adult social associations

During the initial 6 weeks after release into our semi-natural enclosure, we did not detect a difference in social behavior between rearing treatments. Lizards may have been habituating to, and exploring, their novel environment, as well as establishing social structure during this period. Our finding that number and strength of associations (binary and weighted degree) decreased and social heterogeneity (CV of edge weights) increased over this 6-week period supports the hypothesis that lizards were establishing social structure. Consequently, all individuals, independent of the experimental treatment, may have interacted with more conspecifics, and also more frequently, than in a population with an established social structure. The social behavior we recorded over the following 6 months (i.e. lizards sharing refuges overnight) may more clearly reflect this lizard's social preferences than social associations during their daily activity period. During the day, lizard interactions are likely to be influenced by their activity (e.g. encounters with individuals at refuges during foraging or other movements) and may also include social interactions that are short and/or agonistic instead of affiliative. Previous studies have found that crevice-sharing behavior directly reflects tree skink social associations (Bonnnett 1999; Chappele 2003; Duckett et al. 2012), and, because refuges were not limited within our enclosure, sharing a refuge overnight can be expected to reflect an individual's social preference.

We found evidence that rearing treatment affected crevice-sharing behavior over the following 6 months. Isolated lizards were more homogeneous in their social associations, and all lizards, regardless of rearing treatment, became more homogeneous in their social associations over time. At the beginning of the 6-month period, isolated lizards associated more strongly with others, in congruence with our laboratory behavioral assays. But, over time, isolated lizard associations weakened and were with fewer lizards, whereas socially-reared lizard associations were similar in number and frequency over time. Abnormal social rearing environments (e.g. isolation, hand-rearing in captivity, etc.) have been shown to constrain social plasticity in a number of species (Baron and Kish 1960; Tardif et al. 1984; Taborsky et al. 2012; Hesse and Thünken 2014), leading to negative fitness consequences like reduced survival of offspring in 2 species of Callitrichidae (Tardif et al. 1984), as well as reduced longevity, offspring production, and increased infant mortality in some hand-reared species of endangered felids (Hampson and Schwitzer 2016). Yet, in tree skinks, isolation did not constrain flexibility in their social behavior. Isolated lizards, after having experienced a competitive and potentially aggressive social environment, were able to change their crevice-sharing behavior to reflect that of socially-reared lizards. Retaining the ability to plastically respond to changing social contexts may be beneficial for this facultatively social lizard.

Animal behavior has the potential to significantly contribute to conservation biology (Sutherland 1998; Buchholz 2007; Caro 2007). Our study highlights the importance of considering each target species' social system when rearing animals in captivity. In the case of tree skinks, isolation rearing impacted social behavior, but individuals reared in isolation had the highest growth rates (Riley et al. 2017b). Thus, isolation rearing optimizes morphological development but impacts behavioral development. Yet, isolation rearing does not constrain social plasticity. Thus, in the case of tree skinks, a captive-rearing program could benefit from isolation rearing if paired with a "soft-release" or "exposure period" that exposes individual to natural social contexts that could normalize an individual's social behavior prior to release. Conservation programs for Caribbean rock iguanas (*Cyclura* sp.) have utilized this approach: exposing captive-reared juveniles to natural predators and social situations within a controlled environment prior to release (Alberts 2007). An increased understanding of the relationships between social environment and reptile behaviour is also important for ethical considerations about how best to house reptiles within zoos and aquaria (Burghardt and Layne 1995). Overall, knowledge of a target species' social system, and how social rearing environment impacts behavior and fitness, can directly benefit conservation.

Consistency in social phenotypes

Consistency in behavioral traits over time is the antithesis to plasticity. If behavioral traits are consistent across time (i.e. personality; Stamps 2015) it suggests that certain traits are steadily selected for over others. In a dynamic social environment, it is theorized that plasticity in social traits is adaptive (Taborsky and Oliveira 2012). Our findings support this hypothesis, as consistency in the social behaviors we measured was generally not significantly different than what we would expect from chance alone. The exception to this was under laboratory conditions, where tree skinks were within stable social environments for 1.5 years (either social or isolated), and consistency of isolated lizard sociability was moderately repeatable ($R_{adj|age} = 0.427$). In comparison, sociability of socially-reared lizards was not significantly repeatable over the same time frame. When lizards were

released into the much more dynamic environment of a large semi-natural enclosure, consistency in social behavior was lower and did not increase significantly over the following 6 months.

Consistency in social behavior did not differ between rearing treatments. As socially-reared lizards were always exposed to a social environment in our study, we expected their social behavior to be more consistent than isolated lizards (even though there was a change from laboratory to semi-natural conditions for this treatment). In contrast, isolated lizards changed from no social contact during development to a social environment during adulthood. With such a substantial change, we expected isolated lizards to be variable in their social behavior, but only if their behavioral plasticity was not constrained by their development. As both rearing treatments showed similar consistency in behavior, we interpret this finding as evidence that isolation rearing did not affect an individual's social plasticity.

The tree skinks' facultative and variable social system may select for social plasticity. In wild tree skink populations, and those of other *Egernia*-group spp., developing with limited social contact may occur for a proportion of each litter (Bonnett 1999; While et al. 2009b). For example, in White's skink (*Liopholis whitii*) a closely related *Egernia*-group skink, the degree of social contact during development varies depending how related a juvenile is to their social father (While et al. 2009b). Furthermore, tree skinks are long-lived and the social system of this species may be influenced by seasonality and environmental factors (Michael and Cunningham 2010; Duckett et al. 2012). It is likely that throughout a lizard's lifetime it could experience a diversity of social situations ranging from near-isolation to family-living, thus plasticity in social behavior would be adaptive. Such variability contrasts with the more stable social environment of obligate social animals with parental care, which is the basis of the majority of research on this subject (Baron and Kish 1960; Harlow et al. 1965; Varty et al. 2000; Yu et al. 2013). The facultative kin-based sociality of tree skinks, in relation to obligate sociality, best explains our contrasting results and the degree to which these lizards are able to adjust to a novel social environment. Though wild social groupings of tree skinks can differ from what we were examined in our experiment—groups can consist of parent(s) and offspring, and social groups can be larger (i.e. up to 10 individuals in some cases; Chapple 2003). Thus, our understanding of social plasticity, and social environment's impact on tree skink behavioral development, could benefit from further laboratory- and field-based investigations on how parents, kin, and larger group sizes affect social behavior.

CONCLUSION

Isolation rearing affected tree skink social behavior. Isolated, juvenile tree skinks spent laboratory trials closer to an unrelated, adult female. Reflecting the trend we observed in the lab, isolated skinks associated more strongly with conspecifics at the beginning of the 6-month monitoring period within a semi-natural environment. Also, isolated skinks were more homogeneous in the strength of their associations than socially-reared skinks. These findings suggest that isolation rearing resulted in naïve juveniles that were more likely to associate with unfamiliar conspecifics, which could be potentially costly within the tree skink's social system (i.e. a higher chance of infanticide/aggressive encounters).

Although isolation rearing affected social behavior, it did not constrain social plasticity. Isolated lizards gradually decreased the strength and number of associations with conspecifics over

the 6-month monitoring period in the semi-natural environment. We hypothesize that the tree skink's facultative social system selects for plasticity in social behavior, which allows individuals to respond to the variable social contexts they are faced with throughout their lives. Overall, our study demonstrates that the impact social rearing environment has on social behavior may depend on a species' social system, and this finding may have important implications for conservation programs.

SUPPLEMENTARY MATERIAL

Supplementary data are available at *Behavioral Ecology* online.

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Ethical Statement: All protocols in this study were approved by the Macquarie University Animal Ethics Committee (ARA # 2013/039) and work on lizards was approved by the New South Wales National Parks and Wildlife Service, Office of Environment and Heritage (License # SL101264).

Data accessibility: Analyses reported in this article can be reproduced using the data provided by Riley et al. (2018b). The data, and corresponding R code, can also be accessed from the Bitbucket repository at https://julia_riley@bitbucket.org/julia_riley/e-striolata-social-behavior-study.

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